SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Software:Cisco Packet Tracer,Wireshark

Code of Conduct:

*I **T Bhupal / 192425243** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ***bhupalreddy***

Criteria	**Weightage**	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%				
Debugging	25%				
Optimization	20%				
Analysis	20%				
Code Quality	15%				

Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet **192.168.10.0/26**.

Task

Debug the network configuration by:

- Identifying the subnet to which the configured IP address belongs.
- Determining the network address, broadcast address, and valid host range.
- Verifying whether the configured IP address and gateway are valid.
- Identifying the configuration errors.
- Optimizing the configuration by assigning an appropriate IP address and default gateway.
- Calculating the total number of usable host addresses after correction.

2. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network **192.168.1.0/24** into several departments using the subnet masks **/26**, **/27**, and **/28**. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: **2001:db8::ff00:42:8329/64**
- Device B: **2001:db8:0:0:1::1/64**

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Users report that packets destined for **192.168.2.100** are not reaching the intended network.

Task

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

QUESTION 1: Debugging IPv4 Address Configuration

Given Data

- IP Address: 192.168.10.65
- Subnet Mask: 255.255.255.192 (/26)
- Default Gateway: 192.168.10.1
- Expected Subnet: 192.168.10.0/26

Step 1: Identify the Subnet

Subnet Mask = 255.255.255.192

Block Size = 256 − 192 = 64

The subnet ranges are:

1. 192.168.10.0 – 192.168.10.63

2. 192.168.10.64 – 192.168.10.127
3. 192.168.10.128 – 192.168.10.191
4. 192.168.10.192 – 192.168.10.255

The configured IP address 192.168.10.65 belongs to the subnet **192.168.10.64/26**.

Step 2: Calculate Network Details

Host Bits = 32 − 26 = 6

Total IP Addresses = 2^6 = 64

Usable Hosts = 64 − 2 = 62

- Network Address: 192.168.10.64
- Broadcast Address: 192.168.10.127
- Valid Host Range: 192.168.10.65 – 192.168.10.126
- Total Usable Hosts: 62

Step 3: Verify the IP Address and Gateway

The workstation IP address 192.168.10.65 is valid for the subnet 192.168.10.64/26.

However, the configured gateway 192.168.10.1 belongs to the subnet 192.168.10.0/26.

Therefore, the workstation and gateway belong to different subnets.

Identified Error: The default gateway does not belong to the workstation's configured subnet.

Step 4: Optimize the Configuration

Since the workstation must communicate within the intended subnet 192.168.10.0/26, assign the following configuration:

- IP Address: 192.168.10.2
- Subnet Mask: 255.255.255.192
- Default Gateway: 192.168.10.1
- Network Address: 192.168.10.0
- Broadcast Address: 192.168.10.63
- Valid Host Range: 192.168.10.1 – 192.168.10.62
- Usable Host Addresses: 62

Final Answer

Parameter	Corrected Configuration
IP Address	192.168.10.2
Subnet Mask	255.255.255.192
Network Address	192.168.10.0
Broadcast Address	192.168.10.63
Valid Host Range	192.168.10.1 – 192.168.10.62
Default Gateway	192.168.10.1
Usable Hosts	62

Conclusion: The communication failure is caused by a subnet mismatch between the workstation and its default gateway. Assigning 192.168.10.2/26 with gateway 192.168.10.1 corrects the configuration, provided the gateway interface is operational.

QUESTION 2: Debugging IPv6 Address Configuration

Given Data

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Step 1: Expand Both IPv6 Addresses

An IPv6 address contains eight groups of four hexadecimal digits.

Device A

Compressed Address:

2001:db8::ff00:42:8329/64

Expanded Address:

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B

Compressed Address:

2001:db8:0:0:1::1/64

Expanded Address:

2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Determine the Network Prefix

The prefix length is /64. Therefore, the first four hexadecimal groups represent the network prefix.

Device A

Network Prefix: 2001:db8:0:0::/64

Device B

Network Prefix: 2001:db8:0:0::/64

Step 3: Verify Subnet Membership

The first four hexadecimal groups of both addresses are identical:

2001:0db8:0000:0000

Therefore, both devices belong to the same subnet:

2001:db8:0:0::/64

Result: There is no subnet-prefix mismatch in the given IPv6 configuration.

Step 4: Identify Possible Causes of Communication Failure

Since both devices belong to the same subnet, investigate the following possible issues:

1. Network interfaces are down or incorrectly configured.
2. Physical connectivity or VLAN configuration is incorrect.
3. Firewall rules are blocking communication.
4. Duplicate IPv6 addresses exist.
5. IPv6 Neighbor Discovery is failing.
6. Incorrect link-local addressing or interface settings are present.

Step 5: Optimize the IPv6 Configuration

Example of a clear, consistent configuration using the same /64 subnet:

Device A

- IPv6 Address: 2001:db8:0:0::10/64
- Network Prefix: 2001:db8:0:0::/64

Device B

- IPv6 Address: 2001:db8:0:0::20/64
- Network Prefix: 2001:db8:0:0::/64

Both addresses are unique and belong to the same subnet.

Step 6: Calculate the Number of Available Addresses

Total IPv6 Bits = 128

Network Prefix Bits = 64

Interface Identifier Bits = 128 − 64 = 64

Total Possible Interface Identifier Values = 2^{64}

= **18,446,744,073,709,551,616**

IPv6 does not use broadcast addresses. This is the theoretical number of possible 64-bit interface identifier values.

Final Answer

Parameter	**Device A**	**Device B**
Original Address	2001:db8::ff00:42:8329/64	2001:db8:0:0:1::1/64
Network Prefix	2001:db8:0:0::/64	2001:db8:0:0::/64
Subnet Membership	Same subnet	Same subnet
Example Optimized Address	2001:db8:0:0::10/64	2001:db8:0:0::20/64

Conclusion: Both devices already belong to the same IPv6 /64 subnet. The administrator should investigate connectivity, interface configuration, VLANs, firewall rules, and Neighbor Discovery rather than changing the subnet prefix unnecessarily.

QUESTION 3: Analysis and Optimization of IPv4 Subnet Allocation

Given Data

- Original Network Block: 192.168.1.0/24
- Subnet Masks: /26, /27, and /28

The /24 network contains 256 IP addresses, ranging from 192.168.1.0 to 192.168.1.255.

Step 1: Calculate Subnet Sizes

Subnet /26

- Host Bits: 32 − 26 = 6
- Total IP Addresses: $2^6 = 64$
- Usable Hosts: 64 − 2 = 62
- Subnet Mask: 255.255.255.192

Subnet /27

- Host Bits: 32 − 27 = 5
- Total IP Addresses: $2^5 = 32$
- Usable Hosts: 32 − 2 = 30
- Subnet Mask: 255.255.255.224

Subnet /28

- Host Bits: 32 − 28 = 4
- Total IP Addresses: $2^4 = 16$
- Usable Hosts: 16 − 2 = 14
- Subnet Mask: 255.255.255.240

Step 2: Determine Network Details

For demonstration, allocate the three subnets sequentially without overlap.

Subnet 1: 192.168.1.0/26

- Network Address: 192.168.1.0
- Valid Host Range: 192.168.1.1 – 192.168.1.62
- Broadcast Address: 192.168.1.63
- Total IP Addresses: 64

- Usable Hosts: 62

Subnet 2: 192.168.1.64/27

- Network Address: 192.168.1.64
- Valid Host Range: 192.168.1.65 – 192.168.1.94
- Broadcast Address: 192.168.1.95
- Total IP Addresses: 32
- Usable Hosts: 30

Subnet 3: 192.168.1.96/28

- Network Address: 192.168.1.96
- Valid Host Range: 192.168.1.97 – 192.168.1.110
- Broadcast Address: 192.168.1.111
- Total IP Addresses: 16
- Usable Hosts: 14

Step 3: Calculate Allocated and Unused Addresses

Total Allocated Addresses:

64 + 32 + 16 = 112

Remaining Unallocated Addresses:

256 − 112 = 144

Unused IP Address Range:

192.168.1.112 – 192.168.1.255

The remaining range contains 144 IP addresses.

Step 4: Identify Inefficient Allocation

A /26 subnet supports 62 usable hosts, a /27 supports 30, and a /28 supports 14.

If a department requires only 10 devices, allocating a /26 subnet would provide considerably more addresses than necessary. A /28 subnet would provide 14 usable host addresses and would be more suitable.

The exact amount of wasted capacity cannot be determined without the actual device requirements of each department.

Step 5: Recommend an Optimized VLSM Scheme

Variable Length Subnet Masking (VLSM) allocates different subnet sizes according to departmental needs.

Recommended steps:

1. Determine the number of devices required by each department.
2. Allocate the largest required subnet first.
3. Assign smaller subnets to departments with fewer devices.
4. Reserve network and broadcast addresses.
5. Prevent overlapping subnet ranges.
6. Keep unused addresses available for future expansion.

Final Answer

Subnet	Network Address	Valid Host Range	Broadcast Address	Usable Hosts
/26	192.168.1.0	192.168.1.1–62	192.168.1.63	62
/27	192.168.1.64	192.168.1.65–94	192.168.1.95	30
/28	192.168.1.96	192.168.1.97–110	192.168.1.111	14

- Total Allocated Addresses: 112
- Total Unallocated Addresses: 144
- Remaining Unused Range: 192.168.1.112 – 192.168.1.255

Conclusion: VLSM improves address utilization by allocating subnet sizes according to actual departmental requirements. The proposed allocation is valid and non-overlapping, but further optimization requires the number of devices in each department.

QUESTION 4: Debugging IPv6 Address Configuration

Given Data

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Step 1: Expand Both IPv6 Addresses

Device A

2001:0db8:0000:0000:0000:ff00:0042:8329/64

Device B

2001:0db8:0000:0000:0001:0000:0000:0001/64

Step 2: Determine the Network Prefix

Device A Network Prefix:

2001:db8:0:0::/64

Device B Network Prefix:

2001:db8:0:0::/64

Both addresses have the same first 64 bits.

Step 3: Verify Subnet Membership

Both devices belong to the same subnet:

2001:db8:0:0::/64

Therefore, the given addresses do not have a subnet-prefix mismatch.

Step 4: Identify Possible Configuration Errors

Possible causes of communication failure include:

1. Incorrect interface configuration.
2. Physical or logical connectivity problems.
3. VLAN mismatch.
4. Firewall restrictions.
5. Duplicate IPv6 addresses.
6. Neighbor Discovery failure.
7. Incorrect routing or interface status.

Step 5: Optimize the Configuration

Example addresses in the same /64 subnet:

- Device A: 2001:db8:0:0::10/64
- Device B: 2001:db8:0:0::20/64
- Common Network Prefix: 2001:db8:0:0::/64

Both devices must have unique addresses and operational interfaces.

Step 6: Calculate Available Addresses

Interface Identifier Bits = 128 − 64 = 64

Total Possible Interface Identifier Values = 2^{64}

= **18,446,744,073,709,551,616**

Final Answer

Parameter	**Result**
Device A Expanded Address	2001:0db8:0000:0000:0000:ff00:0042:8329/64
Device B Expanded Address	2001:0db8:0000:0000:0001:0000:0000:0001/64
Network Prefix	2001:db8:0:0::/64
Subnet Membership	Same subnet
Optimized Device A Address	2001:db8:0:0::10/64
Optimized Device B Address	2001:db8:0:0::20/64
Interface Identifier Values	2^{64}

Conclusion: Both devices already belong to the same IPv6 subnet. The administrator should investigate interface configuration, connectivity, VLANs, firewall rules, and Neighbor Discovery.

QUESTION 5: Debugging and Optimizing a Router's Routing Table

Given Data

Destination Network	**Next Hop**
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
Default Route	10.0.0.1

Destination IP Address: 192.168.2.100

Step 1: Identify the Matching Route

Destination IP Address = 192.168.2.100

The destination belongs to the network 192.168.2.0/24.

Subnet Mask = 255.255.255.0

Network Address = 192.168.2.0

Broadcast Address = 192.168.2.255

Valid Host Range = 192.168.2.1 – 192.168.2.254

Matching Route:

192.168.2.0/24 → 10.0.0.3

Step 2: Determine the Next-Hop Router

The matching routing entry specifies:

Next-Hop Address: 10.0.0.3

The router forwards the packet to this next hop, provided that it is reachable through an appropriate outgoing interface.

Step 3: Verify the Destination IP Address

The destination IP 192.168.2.100 falls within the valid host range of 192.168.2.0/24.

Therefore, the destination address is valid for the intended subnet.

Step 4: Identify Possible Routing Errors

The routing entry itself correctly matches the destination. Possible causes of packet loss include:

1. Next-hop router 10.0.0.3 is unreachable.
2. The outgoing interface is incorrectly configured.
3. The next-hop router has no route to the destination.
4. The destination network has no return route.
5. Firewall or access-control rules block traffic.
6. The destination device is disconnected or incorrectly configured.

The supplied routing table does not establish which of these issues is responsible.

Step 5: Optimize the Routing Table

The routing table should be represented as follows:

Destination Network	Next Hop
192.168.1.0/24	10.0.0.2
192.168.2.0/24	10.0.0.3
0.0.0.0/0	10.0.0.1

Optimization steps:

1. Verify reachability of next-hop routers.
2. Check the outgoing interfaces.
3. Confirm that the destination network is reachable through 10.0.0.3.
4. Verify return routes.
5. Remove incorrect or redundant routes.
6. Use longest-prefix matching to select the most specific route.

Step 6: Identify the Default Route

Default Route:

0.0.0.0/0 → 10.0.0.1

The default route is used when no more specific route matches the destination and the next hop is reachable.

Final Answer

Parameter	Result
Destination IP	192.168.2.100
Destination Network	192.168.2.0/24
Matching Routing Entry	192.168.2.0/24
Selected Next Hop	10.0.0.3
Destination Address Validity	Valid host address
Default Route	0.0.0.0/0 → 10.0.0.1

Conclusion: The router selects 192.168.2.0/24 and forwards the packet to next hop 10.0.0.3. If delivery fails, the administrator should check next-hop reachability, interface configuration, return routes, and filtering rules.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-structured code with proper comments and documentation	Code is mostly clear with minor documentation gaps	Code lacks structure and proper documentation	Poorly written code with no documentation